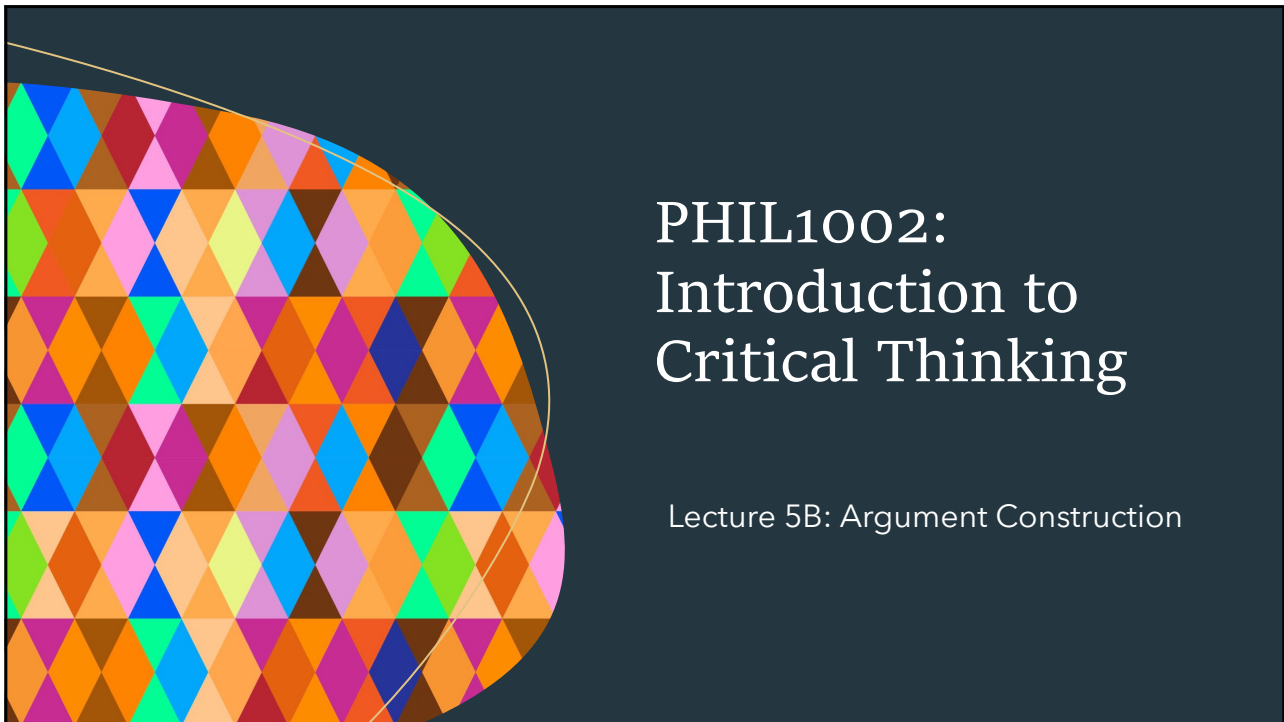




1

Conditionalisation

For every argument, there corresponds a conditional which has as its antecedent the conjunction of the premises and as its consequent the conclusion.

Suppose the premise is P , and the conclusion C , and the argument from P to C :

P Therefore, C

The corresponding conditional in this case would be:

If P then C

2

Conditionalisation

The conditional and the argument are **NOT equivalent** but they **correspond**.

The letter 'P' can stand for as many premises as you like, since a set of premises is equivalent to a conjunction which contains all those premises as conjuncts.

If the argument had been:

[P1 and P2 and P3 and P4], so C

then the corresponding conditional would have been

If (P1 and P2 and P3 and P4), then C

3

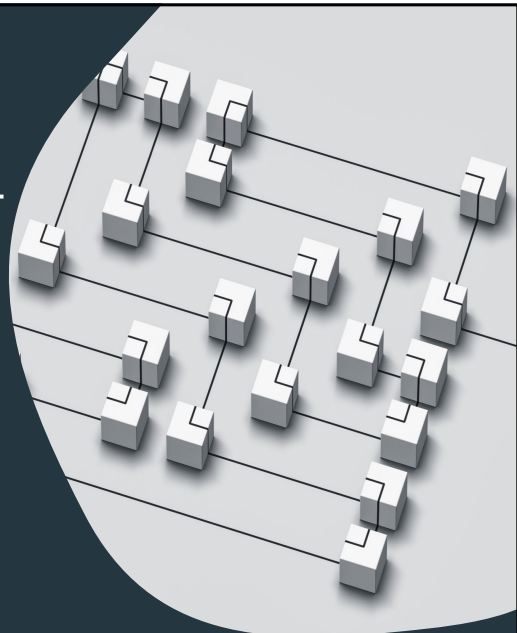
Conditionalisation

When we form the conditional corresponding to an argument, we abandon the assertion of the premises and conclusion.

Instead, we assert that there is a connection between those premises and that conclusion.

That means that if there is such a connection, then the corresponding conditional will be true

If there isn't that connection, then the corresponding conditional will be false.



4

Conditionalisation

When the degree of support between premises and conclusion is at its maximum (valid) then the corresponding conditional is not only true, it is **necessarily** true e.g.:

This ball is red, Therefore, This ball is coloured

Corresponding necessarily true conditional:

If this ball is red, then it is coloured

5

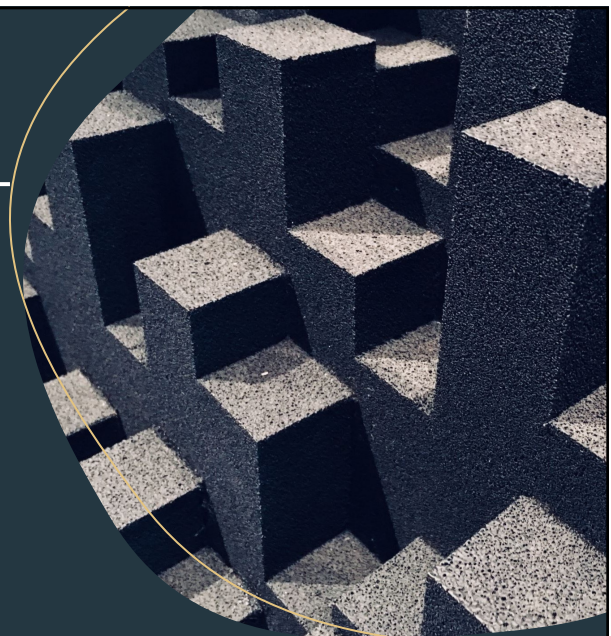
Reductio ad absurdum

Reductio ad absurdum is Latin for 'reduction to absurdity'.

It's a very useful technique for making arguments.

Can be use as a method of refutation.

Can also be used as a method of positive argument.



6



Reductio and conditionalisation

But how does **conditionalisation** fit into this?

First you derive the conditional, that if the premise is true then the conclusion must be true too.

Then the absurdity of the conclusion reflects back on the acceptability of the premise.

7

An example of a *reductio*

- (1) 4 is a prime number [Prem., provisionally assumed]
- (2) A prime number is a number which is not a product of two smaller natural numbers. [Prem., common knowledge]
- (3) 4 is not the product of 2 and 1. [1 & 2, linked]
- (4) If 1 & 2 are true, then 3 is true too. [Conditionalisation]
- (5) It is absurd to think that 4 is not the product of 2 and 1 [Prem. (implicit contradiction)]
- (6) The assumed premise, 1, is false. [4 & 5, Modus Tollens]

8

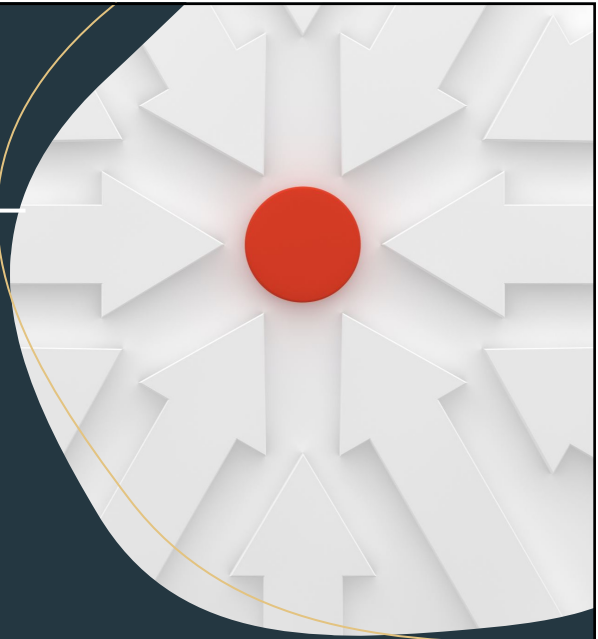
Broad and narrow *reductio*

There is both a broad and a narrow sense of *reductio*.

In a **narrow *reductio*** the argument works by producing a recognisable **contradiction**.

Contradictions are necessarily false.

That allows us to argue back, to the falsehood of the provisionally assumed premise.

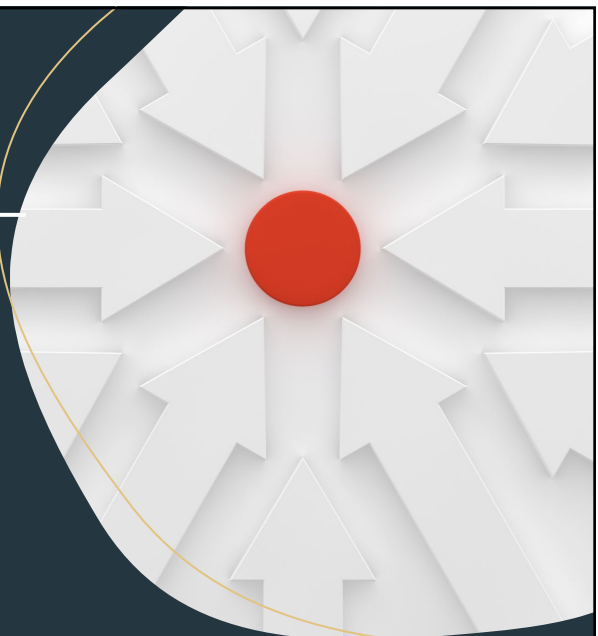


9

Broad and narrow *reductio*

In the **broad** sense, an argument by *reductio* produces a consequence that is thought to be absurd because **obviously false**.

Such a consequence need not be a contradiction, just something that the arguer supposes no one would believe.



10

Broad *reductio*

Reductio in the **broad** sense is less conclusive a refutation than one in the **narrow** sense.

Keep the two senses apart for this reason.



11

Devising a *reductio*

1. P [prem.: provisional assumption]
2. C [1 (some consequence of P)]
3. *If P , then C* [1 & 2, conditionalisation]
4. $\text{Not-}C$ [obvious *ex hypothesi*]
5. $\text{Not-}P$ [3 & 4, *modus tollens*]

You must remember that, in practice, all the difficulty will come in getting from P to C .

12

Ad hominem arguments

Ad hominem, Latin for 'to the man'.

It works not by attacking a statement itself but by attacking the person putting forward that statement.

Here's an example in the second part of this dialogue:

A. *Foxhunting ought to be abolished; it is cruel to the victim and degrading to the participants.*

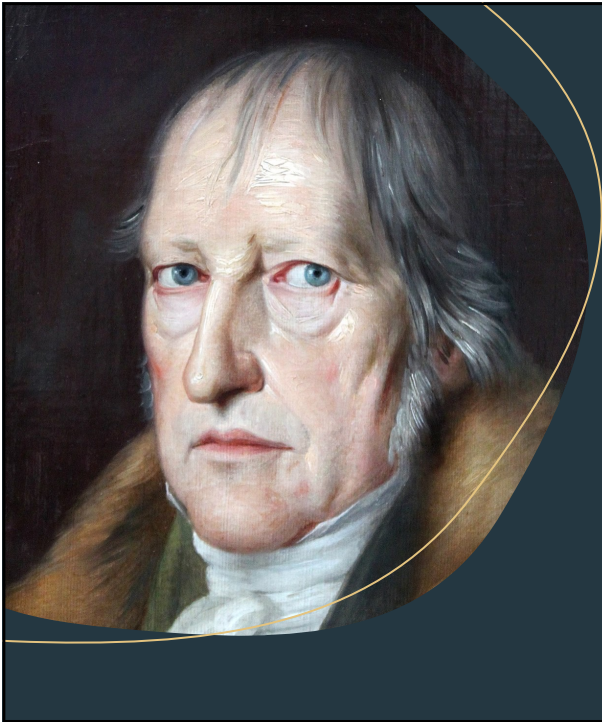
B. *But you eat meat; and I'll bet you've never worried about whether the killing of the animals you eat is cruel to them and degrading to the butchers.*

13



1. *Foxhunting is cruel to victim and degrading to participants.* [prem: accepted by A]
2. *Killing of animals for food is not relevantly different from foxhunting.* [prem: added by B]
3. **Both** *ought to be abolished.* [1 & 2 (if A's original inference correct)]
4. *If 1 & 2 are true, then 3 is true.* [condit.]
5. *A eats meat.* [prem]
6. *A cannot accept 3.* [3 & 5]
7. *A cannot consistently accept 1 and 2.* [4, 6 modus tollens]

14



Ad hominem arguments

You needn't take any notice of Hegel's views on women, because he had an unsatisfactory relationship with his mother and that made him funny about women all his life.

15



Ad hominem arguments

The real target of an *ad hominem* argument is some aspect of a person's mind, character or capacities.

For that reason they can be particularly valuable in the courts.

Here one may wish to discredit a witness's testimony by attacking the person.

For instance, by showing that he is too short-sighted to have reliably identified the defendant, or he's a notorious liar or whatever.

16

Critical thinking...

...or just plain critical!



17

A response...

First, you can use argument forms like *reductio* in a positive as well as a negative way: proving the positive claim by refuting its denial.

Second, these critical skills which we have tried to help you acquire for yourselves are not just to be turned on other people—they are to be used on yourselves, to improve your own arguments.

Third, you're probably already constructing your own arguments.

18

Critical thinking

Argument is a matter of dialogue. Whether you start out on the constructive side, or on the critical side, after a couple of exchanges both participants are in the same boat.

Being a competent critic of the arguments of others, and arguing clearly and effectively yourself, are not, in the long run, different abilities.



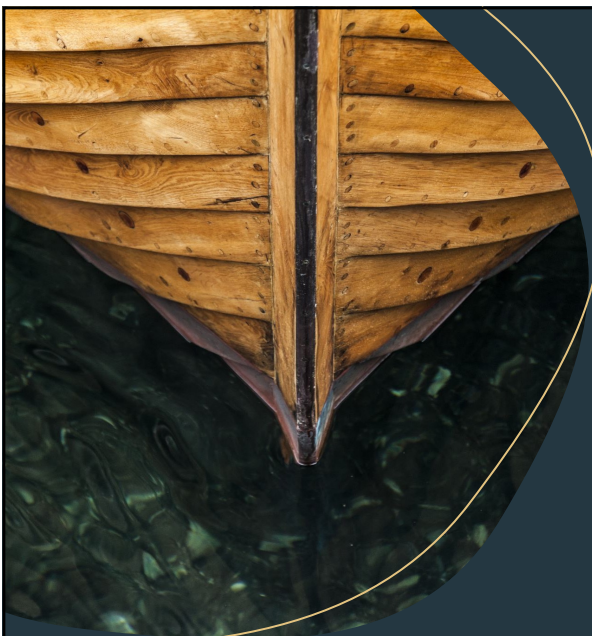
19

Tips: Constructing an argument

Oddly enough, the best place to start is usually with the conclusion.

This is not to say that you should arrive at your conclusions at random, and then just make up arguments to try to rationalise them.

You may have to modify or even abandon the conclusion you start with.



20

Tips: Constructing an argument

Start by formulating the conclusion you think you want to defend.

If it's a generalisation, test it by trying to think of a counter-example to it.

If you can't think of one, then set about trying to build your argument.

If you can, then your conclusion will have to be *at least* modified to accommodate the counterexample.



21

Tips: Constructing an argument

Suppose, for example, you have an essay to write in the philosophy of religion,

and the topic is:

'Does the problem of evil show that God does not exist?'



22

Tips: Constructing an argument

Here is a wrong way to begin such an essay:

Philosophers have argued about the problem of evil for centuries. Some of them have written whole books on the subject, but no consensus has been reached, and the debate looks set to continue.

Here is another wrong way:

First, we must define our terms.



23

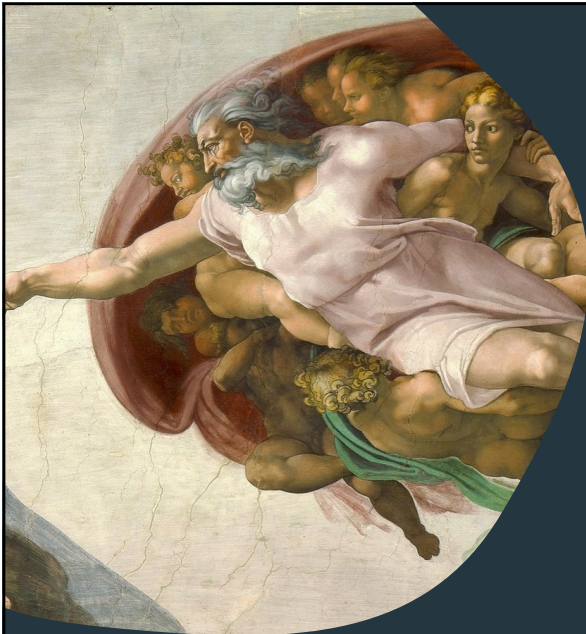
Tips: Constructing an argument

And here is **one** right way:

In this essay I shall contend, first, that there is no problem of evil, and that it can accordingly show nothing about the existence of God; and secondly, that even if there were such a problem, it would be at best inconclusive evidence against God's existence.



24



Tips: Constructing an argument

Or alternatively:

I shall argue that the problem of evil shows that God, as conceived within the Judæo-Christian tradition, cannot exist, and that any being whose existence is not precluded by the problem would necessarily lack at least one of the essential attributes which that tradition ascribes to God.

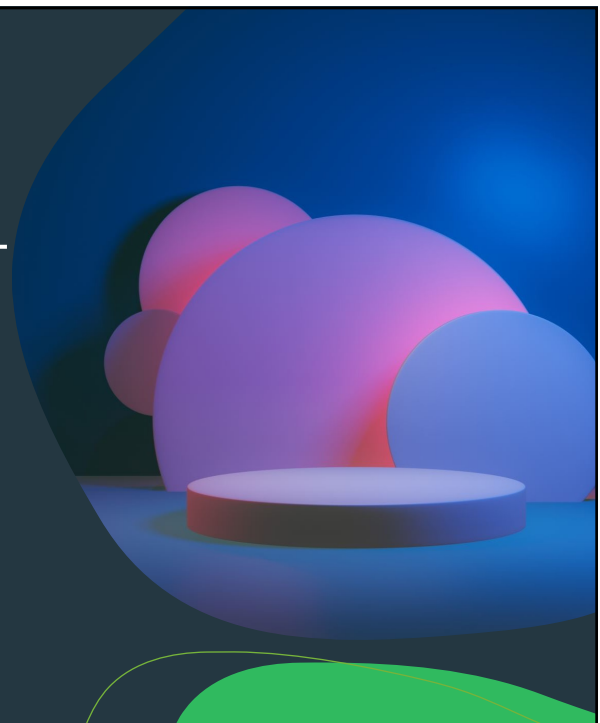
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Tips: Constructing an argument

List every reasonable idea you can think of which would be relevant to proving or disproving that conclusion.

When you have formulated these considerations, select from the list of relevant considerations.

Chose acceptable premises.



26

Tips: Convince the sceptic

Think about to whom your argument is addressed?

Often there will be no particular person, but it helps to imagine the sort of person there might be.

In the context of a controversial topic every controversial claim has to be argued for.



27

Tips: Narrow it down

Don't try to cover everything.

Most topics tend to have a breadth and complexity which **resists comprehensive treatment.**

Often, you can present an argument concerning only one aspect of a topic, provided that aspect is sufficiently central.



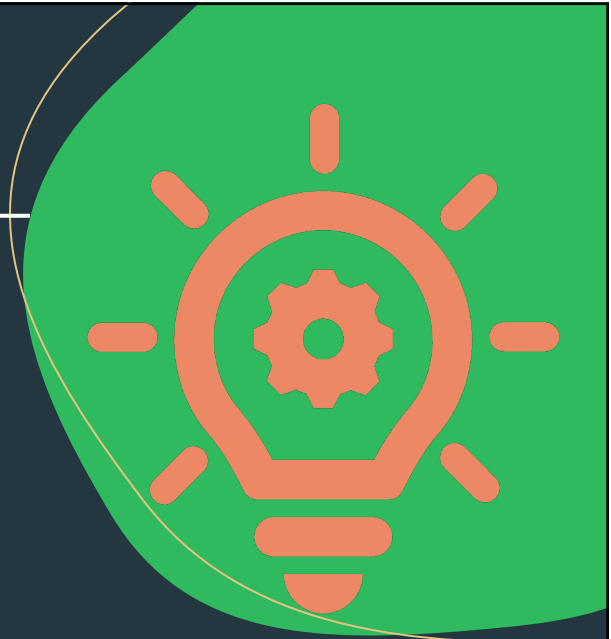
28

Tips: Write clearly and simply

Make your argument it as clear as possible.

First, the actual language you use should be clear and as simple as the subject will allow.

You **may** need to define key terms, if they are being used in some special or technical sense.



29

Tips: Beware of definitions

Definitions are often more obscure than the term they are supposed to be defining.

Don't define the key terms in such a way as to ensure that the argument works.

Premises can be unacceptable by being question-begging—the same is true of definitions.

Definitions don't usually resolve arguments.



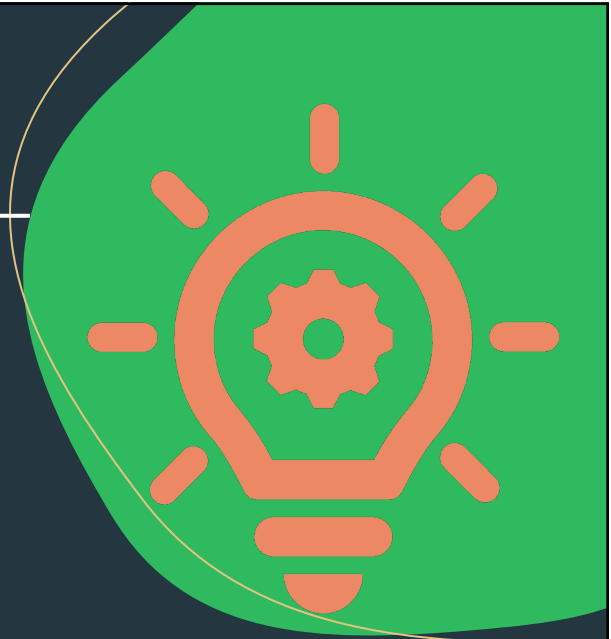
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Tips: Write clearly and simply

Repeat nouns to avoid the sort of ambiguity that arises from using pronouns.

Keep your argument always moving in the same direction.

Some inference indicators precede premises, while others follow them—don't mix them up.



31

An example

It is a fact that (1) <people cannot keep their best New Year's resolutions unless they change their environment to make it more favourable to good behaviour>. For this reason, (2) <it is hopeless to try to get people to behave decently towards one another merely by throwing moral exhortations at them>. Another reason for this is that (3) <virtue is an affair of habits, acquired by doing, and not by listening>. So (4) <we need fewer self-appointed moralists and more social scientists to get at the roots of evil>.

The actual structure of this argument is not clear even when one has put in the numbers and brackets as I have.

(Avoid writing in after thoughts!)

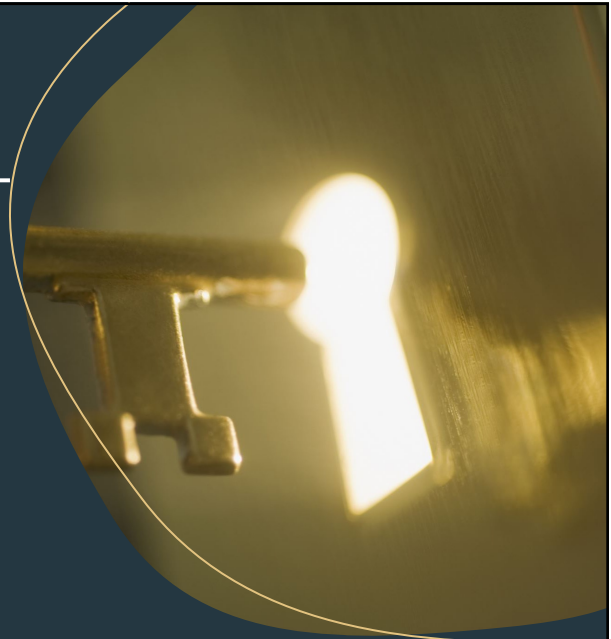
32

Tips: Be consistent

It's easier than you might think to contradict yourself by accident.

This can happen in quite subtle ways.

You may give several different reasons for your conclusion and not notice that two of them proceed from inconsistent assumptions.



33

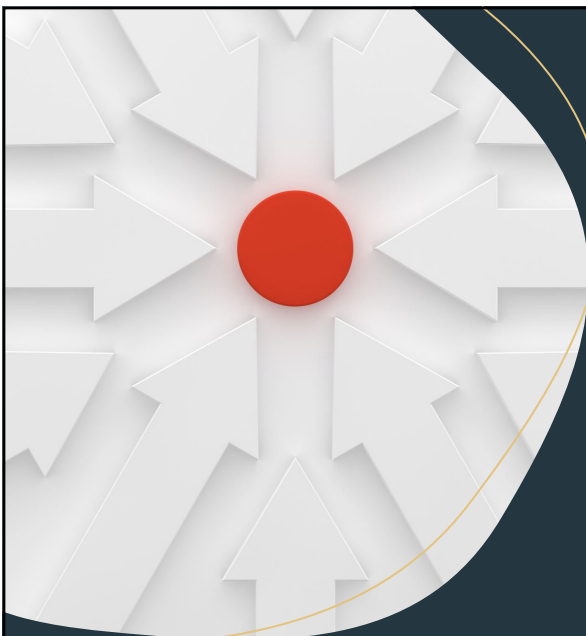
Tips: Present your opponent's view fairly

You can discuss both sides of an issue.

Always give a serious and sympathetic presentation of the opposing point of view.

This may lead to a modification and improvement of your own position.

It demonstrates that your argument is objective and unbiased.



34



Tips: A balanced approach

You may find yourself constructing arguments for both sides and end up unable to choose between them for strength.

This is not a defect, and it is different from merely being inconsistent

It is a sign of honesty.

35



Tips: Anticipating objections

Anticipate objections to your argument.

Think yourself into the role of opponent and critic and put forward the most likely objections to what you have to say.

Then show how they can be met.

36

Constructing and criticising

Criticising arguments is much the same as constructing arguments—the abilities involved do not differ greatly.

You have to be prepared to think of objections and respond to them.

The main difference is in the starting point.

With argument criticism, you begin by criticising someone else.

With argument construction, you do your own criticism both in sharpening up your argument for presentation, and in anticipating the objections to it.

37

Thanks!

Good luck with the rest of the semester!

Please contact me if you have questions or need help.

38